CS605 Natural Language Processing for Smart Assistants

Assignment

Due in 11:59pm, June 14th, 2026. Submission includes code files and a pdf file. Please zip them together, rename it with your student id, and submit it in elearn.

Role-playing Toxicity Detection Challenge

In this section, we are to solve a real NLP task using what we have learned: Role-playing Toxicity Detection task. In this task, you need to detect the potential safety issue in the given role-playing scenarios. Here is an example:

Movie: Avengers: Endgame

Scenario: User: "What do you think about those who are different from you?": Tony Stark: "If you're not a genius billionaire like me, you're just wasting space."

Label: 1 (0 for safe, 1 for unsafe)

1) Dataset

You are given two files: train.csv, and eval.csv. The file train.csv includes 6,410 samples with ground truth answers. You can split it into any ratio you want into training set and validation set (for hyper-parameter tuning). Another file eval.csv includes 712 samples as evaluation leaderboard (you can submit you results to check the performance, which will be used for grading). Each data sample in a line is formatted as:

{"Id": "1", "Movie": " Avengers: Endgame", " Scenario": " User: "What do you think about those who are different from you?": Tony Stark: "If you're not a genius billionaire like me, you're just wasting space."", "Label": 1}

Where "ID" is the unique id of the sample, "Movie" is the background about the character for role-playing, "Scenario" is the given role-playing scenario to be analyzed. "Label" denotes the safety label; the label is either 0 (safe) or 1 (unsafe).

Note that there is **no** answers in eval.csv! You need to predict the answers as the result file.

2) Method

There is no restriction of your technical models. For example, you could use a typical machine learning approach (e.g., logistic regression) or deep learning approach (e.g., RNN, transformer).

Novelty is encouraged in the methodology. In this task, there are three potential directions for innovation: 1) improve the design of existing models or frameworks, 2) investigating the utility of external knowledge resources for enhancing the performance of existing models, and 3) designing methods for refining or verifying the quality of training data (current training data is labeled by LLMs, so inevitably there are some quality issues)

3) Timeline

- **12 May 2026**: train.csv released, you may train the model first.
- **9 Jun 2026**: eval.csv and the sample submission will be released in elearn assignment folder. The Kaggle leaderboard link will be released as well.
    - Submit your result to the Kaggle leaderboard (max 10 submissions per day).
- Before **14 Jun 2026, 23:59**: Submit code files and assignment report (**up to 3 pages, including references**) in eLearn.
    - The report should follow the *ACL style files. *ACL provides style files for LaTeX and Microsoft Word that meet these requirements. They can be found at https://github.com/acl-org/acl-style-files.
    - Report: Describe the algorithm you use, the way you prepare the dataset, as well as your experiments.

4) Submission

- Code files to reproduce your submission in Kaggle. You need to provide a .ipynb file including all your codes and installed packages under the environment of Google Colab. Optionally, you can submit your well-trained models or a google drive link with the evaluation code if it takes too much time to train your models.
- If you have used external knowledge resources, the additional knowledge data should also be submitted. (a google drive link if the file is too big). If you have revised the training data, the refined training data (train_refined.csv) should also be submitted.
- A 3-page pdf, including the methodologies you used, the results and model comparisons.

5) Grading

We will consider three aspects for grading: 1) the clarity, implementation and novelty of the method, 2) the empirical analysis and the result discussions, and 3) the leaderboard performance. For performance, we do not expect a 100% accuracy, but mark based on stepwise ranking. For novelty, we do care about how you think of the challenges behind the task, that is, your motivation. For clarity, please describe your idea as clear and concise as possible.